OverFlow: A Geospatial Decision Support System for Flood Evacuation Planning

CAPSTONE PROJECT

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ABSTRACT

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1 Introduction

Flood-evacuation planning requires decisions about where people can go, which routes they can use, and whether available facilities and transport can accommodate them. These decisions are interdependent: closing a road may make a nearby evacuation center unreachable, while reducing a center's capacity may require residents to travel farther or leave some demand unserved. Examining these conditions together can help planners identify limitations in a proposed evacuation arrangement before it is needed.

This study proposes *OverFlow: A Geospatial Decision Support System for Flood Evacuation Planning* for Laoag City, Ilocos Norte. It brings geographic information and configurable evacuation conditions into an interactive three-dimensional (3D) environment. The intended users are local disaster planners and responders who need to examine and compare possible evacuation arrangements. The system is designed to support their judgment through scenario analysis and clearly stated planning assumptions.

1.1 Project Context

Geographic information provides a foundation for evacuation planning by locating communities, roads, bridges, hazard areas, and potential destinations. However, viewing these features does not by itself determine whether a route remains usable under a particular closure, whether a destination has sufficient capacity, or whether available vehicles can serve the expected evacuees. These questions require geographic information to be considered alongside the conditions of a specific planning scenario.

Laoag City is the selected study area. The research team has received a City Hall data package containing terrain, barangay boundaries, population figures, roads, flood-

susceptibility information, and other geographic references. These materials provide a starting point for developing the local planning environment. Their receipt does not establish that every layer is sufficiently current, complete, or accurate for the intended analysis. A usable evacuation-center register and current rescue-resource information are still needed, and the pilot barangays and operating organization remain to be identified.

The available information also requires distinctions that affect how a scenario is interpreted. Population records describe the residents covered by a source; they do not directly establish how many people will need evacuation in a particular event. Likewise, a static flood-susceptibility map provides reference information about hazard conditions but does not show the observed extent of a specific flood or the outcome of a newly configured rainfall scenario. Treating these records according to their actual meaning is necessary for credible planning comparisons.

The project addresses the need for a workflow that connects these geographic references with editable planning conditions. A planner should be able to select barangays, specify expected evacuees, change road access and center availability, and examine the resulting routes, assignments, and resource requirements. When connectivity or capacity is insufficient, the system should make the remaining demand visible. Comparing such scenarios can help users examine where a proposed plan depends on a particular road, center, or transport assumption.

1.2 Purpose and Description

The purpose of *OverFlow* is to support the preparation and review of flood-evacuation arrangements in Laoag City through configurable scenarios. The planned interface allows users to select one or more mapped barangays and inspect available terrain, road, bridge,

evacuation-center, and hazard-reference information in a 3D scene or an overhead view.

For each selected barangay, the planner views the source population figure as a reference and enters the expected evacuees separately, either as a number or as a percentage of that reference. The planner also specifies rainfall intensity and duration, manual road or bridge closures, evacuation-center status and usable capacity, available vehicles and their capacities, and an assumed travel speed. These inputs define the conditions under which an evacuation arrangement is examined.

The evacuation-analysis component is intended to identify feasible and alternative routes through the available road network and assign evacuees to reachable centers within their usable capacities. Its outputs include routes, center assignments, capacity use, estimated travel times and vehicle requirements, and the number of evacuees who cannot be assigned under the configured conditions. Explanations of insufficient capacity or unavailable connections help the planner identify which assumptions or resources may need to change.

Users are intended to save, reopen, duplicate, and compare scenarios, as well as export summaries for planning discussions. This supports examination of how changes in demand, road access, shelter availability, and transport assumptions affect the resulting arrangements. Terrain-based flood classification is also planned, but its rainfall rule and evaluation remain to be established. Its development and limitations are distinct from the route and capacity analysis.

1.3 Objectives of the Study

The general objective of this study is to develop and evaluate *OverFlow*, a 3D geospatial decision-support system for configuring and comparing flood-evacuation planning scenarios in Laoag City, Ilocos Norte.

Specifically, the study aims to:

1. assess and integrate available terrain, barangay, road, hazard-reference, population, evacuation-center, and resource information for the selected Laoag City pilot area;
2. provide barangay-based scenario configuration that distinguishes population references from expected evacuees and allows changes to rainfall assumptions, road and bridge access, center availability and capacity, and transport resources;
3. develop and evaluate a terrain-based flood-classification approach using elevation and rainfall inputs, with its assumptions and limits explicitly stated;
4. analyze feasible and alternative routes and reachable-center assignments subject to capacity constraints, estimate travel times and vehicle requirements, and report unserved demand;
5. present geographic information and scenario results through interactive 3D and overhead views, with facilities for saving, comparing, and exporting scenario summaries; and
6. evaluate the developed system's functionality and support for examining flood-evacuation planning scenarios.

1.4 Significance of the Study

For local disaster planners and responders, *OverFlow* is intended to provide a common environment for examining how road access, evacuation demand, center capacity, and transport availability affect a proposed plan. Explicitly reporting unserved demand may help users recognize arrangements that require additional capacity, another reachable destination,

or a revised transport assumption. Saved scenarios and exported summaries may also support discussion and review of alternatives.

For local engineering and planning personnel, barangay responders, and evacuation-center managers, the system may provide a shared geographic reference for discussing road disruptions, origin areas, destination availability, and capacity requirements. Its usefulness will depend on the quality of the underlying records and the extent to which the scenario assumptions reflect the planning situation being considered.

Community residents are intended indirect beneficiaries of more informed preparedness planning. Any improvement in actual evacuation outcomes would require evidence beyond the development of the software. For researchers and developers, the study may provide a documented example of integrating local geographic data, constrained evacuation analysis, and scenario comparison, including the limitations encountered when source information is incomplete.

1.5 Scope and Delimitations of the Study

The study covers the development and evaluation of a 3D geospatial flood-evacuation planning system for Laoag City, Ilocos Norte. Analysis will be organized around one or more selected barangays. The pilot barangays, implemented geographic coverage, operating organization, and deployment arrangement remain to be finalized. The scope includes scenario configuration, geographic inspection, feasible-route analysis, capacity-constrained center assignment, travel and vehicle estimates, reporting of unserved demand, and the storage, comparison, and export of scenario summaries.

Source population figures will serve as reference values, while expected evacuees will be entered for each scenario. Center capacity and vehicle availability will likewise represent the

values specified for that scenario. Estimates will depend on stated travel-speed, capacity, and scheduling assumptions. The system will report infeasible assignments rather than assume that every evacuee can reach a center with sufficient space. Exact route costs, alternative-route selection, allocation criteria, and time-estimation procedures remain part of the methodology to be specified and evaluated.

The planned terrain component uses a digital elevation model (DEM), D8 flow direction, and flow accumulation to describe terrain drainage paths and contributing area. Converting rainfall inputs into a flood-classification indicator still requires a defined rule, threshold, and evaluation. These terrain operations alone do not establish flood depth or timing. The supplied flood-susceptibility map will be treated as a reference layer, and drainage infrastructure will be displayed only as geographic context. Any automatic restriction of roads based on computed flood conditions will require a separately defined and evaluated rule.

The received City Hall data supports initial development, but its suitability for the final analysis remains subject to checks on coverage, consistency, currency, accuracy, and permitted use. The supplied elevation grid is coarse and cannot establish detailed street-level flood behavior. Access to suitable finer terrain data has not yet been secured. Evacuation-center, current resource, and event-specific flood records remain incomplete. Consequently, implemented coverage and analytical claims will be limited by the data and evidence available during the study.

The study does not cover real-time flood forecasting, public navigation, emergency dispatch, or direct control of physical infrastructure. The planned flood component is not a calibrated hydraulic model and does not represent detailed infiltration, drainage-network hydraulics, or flow velocity. System outputs are intended as planning estimates

whose usefulness and limitations must be evaluated using relevant local data and scenarios; they do not establish actual road safety or guarantee successful evacuation. Evaluation participants, baselines, instruments, and acceptance criteria remain to be specified in the methodology.

1.6 Operational Definition of Terms

D8 Flow Direction. Refers to the planned terrain operation that assigns each elevation-grid cell a flow direction toward its steepest downhill neighbor among eight adjacent cells, where such a neighbor exists.

Decision-Support System. Refers to a system that organizes information and analyzes alternatives to assist a human decision-maker. In this study, it supports the examination and comparison of evacuation scenarios.

Digital Elevation Model (DEM). Refers to a grid of elevation values used to construct the terrain view and support terrain analysis in *OverFlow*.

Evacuation-Center Allocation. Refers to the assignment of expected evacuees to reachable centers without exceeding the usable capacities specified for a scenario.

Evacuation Demand. Refers to the expected number of evacuees entered by the planner for a selected barangay. It may be specified directly or as a percentage of the source population reference.

Feasible Evacuation Route. Refers to a path through the road network that connects an origin to an evacuation center under the access conditions represented in a scenario.

Flood-Evacuation Scenario. Refers to a configured set of rainfall assumptions, selected barangays, evacuation demand, road and bridge access, center availability and capacity, and transport assumptions used to examine an evacuation arrangement.

Flood Susceptibility. Refers to the susceptibility classification shown in the supplied geographic reference layer. It is distinct from observed flooding during a particular event and from a scenario-specific computed result.

Flow Accumulation. Refers to the count or area of cells contributing flow to a cell along the derived terrain flow directions. It does not by itself represent water depth.

OverFlow. Refers to the proposed 3D geospatial decision-support system for flood-evacuation planning in Laoag City developed in this study.

Population Reference. Refers to a source population figure displayed for a barangay when configuring a scenario. It describes residents covered by the source and serves as a baseline for setting evacuation demand.

Unserved Demand. Refers to expected evacuees who cannot be assigned to a reachable center within the available capacity under the configured scenario conditions.

Usable Capacity. Refers to the number of evacuees that a center is allowed to accommodate in a scenario, as specified by the planner.

2 Review of Related Literature

2.1 Introduction

This review examines the studies and systems that provide the main research and methodological foundations for a geospatial decision-support system for flood evacuation planning. The reviewed works address four connected concerns: the effect of flooding on road accessibility, the assignment of affected populations to evacuation centers, the simulation and comparison of evacuation scenarios, and the use of terrain and behavioral information in evacuation analysis. Recent studies from 2022 to 2026 are emphasized to reflect current geospatial and optimization approaches. Earlier works are retained where they establish a foundational method, document an influential related system, or provide locally relevant evidence that remains directly applicable to the study.

Taken together, the literature shows that flood evacuation planning is not adequately represented by a static map or a shortest route calculated under normal road conditions. A useful planning system must relate the flood condition to the road network, determine which destinations remain reachable, respect shelter capacity and other operational constraints, and allow planners to compare the consequences of different assumptions. These requirements correspond to the objectives of OverFlow: to integrate terrain, road, population, evacuation-center, and resource information; identify feasible routes and center allocations; and present alternative scenarios through an interactive geospatial interface.

2.2 Flood-Aware Road Accessibility and Geospatial Decision Support

Li et al. synthesized 211 publications and technical documents on urban flood evacuation, organizing the field around flood simulation, hazard assessment, shelter identification, route development, and the movement of pedestrians and vehicles [1]. Their comparison covered 23 experimental studies of instability in floodwater, 20 route-planning algorithms, and seven evacuee-movement models. The review found that metaheuristic algorithms are effective for complex evacuation routing and identified movement speed, evacuee psychology, multimodal evacuation, and overhead power-line hazards as areas requiring further investigation.

This synthesis provides a useful structure for evaluating the narrower studies in this review because it shows that routing, shelter choice, and movement assumptions depend on the preceding hazard assessment. It also cautions against treating any single algorithm as a complete evacuation plan. For OverFlow, the review supports a workflow in which the terrain-based hazard representation constrains the road network before routes and center allocations are compared. The breadth of Li et al.'s evidence does not validate OverFlow's specific D8 thresholds or local travel assumptions, which still require separate definition and testing.

Alabbad et al. developed a web-based decision support framework that evaluates road and bridge accessibility under multiple flood conditions and provides shortest-path, facility-allocation, and service-area analyses [2]. Its tests in Cedar Rapids and Charles City showed that inundation can disconnect parts of a road network, lengthen routes, and change which critical facility is nearest and reachable. The study is particularly relevant because it treats flood extent as an operational constraint on a graph rather than as a visual overlay alone.

It also demonstrates that flood-related network analysis can be delivered through a web interface to users who do not operate specialist GIS software.

The framework supports OverFlow's proposed integration of flood-affected areas with road routing and its use of an accessible map-based interface. Its scope, however, is centered on transportation accessibility and emergency-facility placement. It does not provide a complete evacuation-planning model that jointly accounts for affected populations, shelter capacities, available resources, and estimated evacuation time. Its use of prepared inundation maps and two Iowa case areas also means that its results cannot be transferred directly to a Philippine locality without locally appropriate terrain, road, and population data. OverFlow can adopt the study's flood-constrained graph principle while evaluating it within a localized planning workflow.

Suwanno et al. combined GIS-based flood susceptibility analysis with road-network analysis to identify suitable evacuation areas and routes in Nakhon Si Thammarat, Thailand [3]. Elevation, slope, land use, proximity to water, and other spatial criteria were used to locate flood-prone areas and assess evacuation alternatives under normal and flooded conditions. Because the case is situated in Southeast Asia, it provides a closer geographic and environmental comparison than studies based only in North America. It also confirms that the safety of a destination and the accessibility of its connecting route must be evaluated together.

The study offers strong support for using GIS layers to screen evacuation centers and calculate routes, but its result is principally a set of suitable areas and paths derived from the selected criteria and weights. Such an approach is valuable for preparedness, yet it provides limited support for repeated what-if comparisons involving road closures, changing demand, center status, and resource availability. For OverFlow, the study informs the selection and

spatial validation of routes and centers, while the proposed scenario mechanism extends the analysis by allowing planners to change operational conditions and compare the resulting plans.

Parajuli et al. similarly combined flood-susceptibility mapping with evacuation-route analysis in Siraha Municipality, Nepal [4]. They used a GIS-based Analytical Hierarchy Process to combine nine flood-conditioning factors, then applied closest-facility and service-area analyses to connect assembly points with shelters. Their route costs accounted for the effects of slope and flooding on pedestrian speed, while a flood map derived from 2019 synthetic-aperture-radar imagery was used to check the susceptibility result. The study classified approximately 48.6% of the municipality as flood susceptible, placed 22 candidate shelters in the safe zone, and found that 12 evacuation routes could be completed within 30 minutes. It also reported that 74.03% of the historically inundated area fell within zones classified as highly susceptible.

The study strengthens the methodological case for validating a hazard classification before using it to constrain evacuation routes. It also demonstrates that travel time can be more informative than distance when terrain and flood conditions reduce movement speed. However, its AHP result depends on expert-assigned criteria and secondary spatial data, and the authors identify damaged roads, collapsed bridges, changing flood depth, and different walking abilities as matters for further work. OverFlow can use the same general sequence of hazard screening followed by network analysis, but its D8-based classification and route estimates require separate local validation rather than adoption of the Nepalese criteria or reported travel times.

Bolanio et al. provide a Philippine example that joins evacuation-center assessment with web-based geospatial access [5]. In Las Nieves, Agusan del Norte, the researchers

used weighted overlay and multicriteria analysis to assess slope, elevation, land cover, road access, proximity to water and electrical sources, and proximity to fuel stations. They also evaluated 11 existing evacuation centers against a 100-year flood layer and developed a web mapping application for center status, spatial capacity, and other flood-related information. The suitability analysis classified 85.1% of the municipality as moderately suitable, 0.3% as highly suitable, and 14.6% as having low suitability. Ten existing centers were moderately suitable, while the Maningalao Evacuation Center was considered unsuitable because it was exposed to the modeled flood.

This case is relevant to OverFlow because it shows that a local-government web application can combine spatial suitability, center information, and flood exposure in a Philippine setting. It also supports checking existing centers rather than assuming that every listed facility is safe. Its weighted-overlay output, however, represents suitability under selected criteria and weights; it does not simulate population assignment, route congestion, or changes across repeated operational scenarios. OverFlow can extend this type of local geospatial inventory through scenario comparison, provided that center capacity, status, and flood exposure are confirmed from authoritative local data.

Abuzo et al. evaluated vulnerable routes in Cagayan de Oro City under normal and flood conditions by combining baseline road and traffic data, travel-condition and perception profiles, spatial analysis, and traffic-flow simulation [6]. Their comparison identified changes in road performance and produced maps and mitigation strategies for managing affected routes. One evaluated alternative route had floodwater affecting 60% of its links, illustrating how a route that exists in the ordinary network may lose much of its practical value during a flood.

This Philippine case strengthens the local basis for comparing a baseline road network

with a flood-constrained version instead of reporting a route without a reference condition. It also suggests that evaluation should cover network-wide effects and alternate-route vulnerability, not only whether one origin can still reach one destination. Because the results depend on Cagayan de Oro's road, flood, and travel conditions, they do not establish performance values for OverFlow. They instead support a methodology in which the system records which links become affected, recalculates accessibility, and presents the change between normal and configured flood scenarios.

Ogawa et al. designed a route-search method for dynamic inundation in Obihiro City, Japan, by modifying Dijkstra's algorithm with road weights based on distance from the present flooded area and the flood extent predicted ten minutes later [7]. Across different levee-breach and shelter conditions, the method directed evacuees farther from the expanding hazard and produced more successful simulated evacuations than the conventional route model. In one tested condition, the number of successful evacuees was 2.16 times that of the comparison model.

The study advances flood-aware routing beyond removing links already marked as flooded: future hazard movement also changes the cost of roads that remain open. This is relevant to OverFlow's scenario logic because a currently passable shortest path may lead toward an accumulating hazard. However, the reported improvement depends on a time-varying inundation model and the study's simulation assumptions. OverFlow's terrain-based accumulation cannot claim the same predictive horizon unless hazard progression is implemented and validated; its present contribution is to make hazard exposure visible and allow route results to change when the configured affected area changes.

Jeon et al. developed a map-API-based guidance system that combines predicted water levels, flood-risk extent, user location, shelter and terrain data, and pedestrian routing

through the T-Map application programming interface [8]. In a South Korean testbed near Yeojubo, the prototype identified the user's position, displayed warnings and the expected hazard area, calculated a route to a shelter, and canceled and recalculated that route when flooding affected the path or destination.

The prototype demonstrates a clear interaction pattern for an evacuation interface: the hazard state, current location, destination, and reason for rerouting should be visible together. This supports OverFlow's planned geospatial presentation and recalculation of feasible routes when roads or centers become unavailable. The study verifies that the prototype functions under its test conditions, but it does not provide field evidence that the guidance reduces casualties or remains reliable during an actual flood. OverFlow should likewise distinguish functional system evaluation from validation of hazard accuracy, route safety, and real-world evacuation outcomes.

2.3 Shelter Capacity and Population Allocation

Qiu et al. integrated a two-dimensional shallow-water model, entropy-weighted flood-risk assessment, multi-objective shelter selection, and agent-based evacuation simulation [9]. The model selected 26 shelters that covered the affected population in the study area. In the subsequent simulation, the evacuation rate rose from 97.4% to 98.4%, the modeled death rate fell from 1.94% to 0.98%, and casualties decreased by 133 compared with the original shelter arrangement.

The study strengthens the case for evaluating proposed shelter locations through a separate evacuation model rather than accepting an optimization result without testing. Its reported reductions are simulation outputs for one modeled urban flood and therefore cannot be interpreted as expected outcomes for OverFlow. The applicable methodological lesson

is to keep hazard assessment, allocation, and evaluation distinct: OverFlow should report whether its selected centers cover demand under the configured constraints and validate the resulting assignments against defined scenarios and local expert review.

Sharbaf et al. proposed a risk-based shelter-network design model that integrates population exposure, shelter risk, evacuation risk, pedestrian accessibility, and flood conditions in a single optimization framework [10]. Applied in Haiti, the model seeks to maximize coverage of vulnerable populations while reducing the risks associated with shelter locations and evacuation paths. Its contribution is the recognition that a nearby shelter is not automatically an appropriate assignment: both the facility and the route leading to it may be exposed to flooding. The study therefore provides a current basis for treating evacuation-center allocation as a constrained risk problem rather than a nearest-facility lookup.

This model directly supports OverFlow's objective of assigning affected populations to accessible centers while respecting safety and capacity considerations. Nevertheless, Sharbaf et al. focus on strategic shelter-network design and pedestrian evacuation, whereas OverFlow is intended to help local planners test operational scenarios involving existing centers, road interruptions, and available resources. The difference suggests that OverFlow should not claim to reproduce the study's optimization model unless the same objective function, constraints, and validation procedures are implemented. Instead, the study supplies criteria against which OverFlow's allocation method can be defined and evaluated, including capacity satisfaction, population coverage, and route exposure.

Wang et al. examined pedestrian evacuation planning for a dam-break flood by assessing road difficulty, disaster risk, and pedestrian demand, then applying fuzzy VIKOR and a staged dual-objective planning method to route and shelter decisions [11]. Their Hurricane

Katrina-based New Orleans case identified high-risk roads near Lake Pontchartrain and selected 22 and 18 shelters from 34 candidates in two simulated planning scenarios. The alternative results reflect different trade-offs rather than a single solution that is best under every objective.

The study supports evaluating shelter assignments together with the risks and demand carried by their connecting roads. It also provides a reason for OverFlow to disclose competing measures such as coverage, route exposure, travel cost, and capacity instead of hiding them behind one score. Its dam-break representation, indicator weights, and New Orleans data are specific to the case, so the selected shelters and performance values cannot be transferred to the project. The usable contribution is the multi-criteria planning structure and the comparison of feasible alternatives under stated priorities.

De Veluz et al. developed a scenario-based, multi-objective location-routing model for pre-disaster planning in Quezon Province, Philippines [12]. The study represented 1,242 barangays and 11,868 road links under eight typhoon scenarios, used Project NOAH risk maps and consultation with the provincial disaster-risk office, and sought trade-offs among facility cost, travel time, and the number of vehicles. A multi-objective particle-swarm method generated alternative plans for distribution hubs, evacuation centers, facility capacities, routes, and vehicle requirements and was compared with other optimization approaches.

This work provides locally relevant evidence that evacuation-center placement, road routing, capacity, and logistics should be examined as one scenario-dependent planning problem. Its Pareto solutions also show why planners may need to choose among several defensible plans according to available facilities, vehicles, and acceptable travel time. The province-scale optimization requires risk, demand, capacity, cost, and fleet data that

OverFlow has not yet confirmed. Accordingly, the study guides the system's comparison measures and its presentation of trade-offs, while the project's actual allocation procedure must be limited to the local data and objectives established in the methodology.

Gama et al. formulated a multi-period shelter location-allocation model that determines when shelters should open, when evacuation orders should be issued, and how evacuees should be assigned as flood and road conditions change over time [13]. The model accounts for limited shelter capacities, time-dependent travel conditions, and the gradual availability of facilities. It is older than the preferred five-year review period, but it remains important because it connects the temporal development of a flood with capacity-constrained allocation and demonstrates why a single static assignment may be insufficient.

The authors also report that exact optimization becomes computationally demanding for realistic problem sizes and therefore use a simulated-annealing heuristic. This limitation is relevant to the design of OverFlow: a planning tool must return understandable results within a practical interaction time, even when an exact optimum is expensive to compute. The study supports scenario comparison and time-sensitive allocation, while also indicating that OverFlow must define its problem size, objective function, and acceptable solution time before describing any route or allocation as optimal.

Bera et al. integrated multi-hazard susceptibility mapping with shelter location-allocation for a mountainous community in the Western Ghats of India [14]. Random Forest models implemented through Google Earth Engine were used to map flood and landslide susceptibility, after which schools were treated as candidate shelters and households as demand points. A P-median model sought shelter arrangements that reduce household travel distance, while a maximal-covering model measured how many households could reach a shelter within specified time limits. When the framework recreated the conditions

of a destructive 2005 rainstorm, the existing shelters could not serve all households within either 30 or 60 minutes.

The result shows why shelter planning must consider both hazard exposure and the portion of the population that can actually reach a facility within an operational limit. It complements Sharbaf et al. by offering a GIS-oriented approach that is easier to interpret, although its single-objective location models simplify the competing demands that arise in an actual evacuation. For OverFlow, household or barangay demand, route accessibility, center capacity, and a stated travel-time cutoff should be evaluated together. The study does not establish that a P-median or maximal-covering formulation is automatically appropriate for the project; the chosen allocation rule must reflect available data and the decisions local planners need to make.

Rendana et al. mapped both flood risk and potential shelter suitability in Palembang, Indonesia, using an integrated GIS and Analytical Hierarchy Process approach [15]. Their criteria included elevation, population, slope, land cover, distance from rivers, drainage density, distance from roads and settlements, and soil type. The resulting flood-risk map classified 30.3% of the area as high risk, 60.5% as moderate risk, and 9.2% as low risk. The shelter-suitability assessment was more restrictive: only 4.1% of the area was highly suitable, while 19.4% was moderately suitable, 60.5% had low suitability, and 16.1% had very low suitability. The best locations were concentrated in higher northern and northwestern areas farther from the river.

These findings distinguish a low-risk location from a practically suitable shelter location: road access, proximity to settlements, terrain, and supporting conditions still affect whether the site can serve evacuees. This distinction is useful for OverFlow's separation of hazard screening from center allocation. Nevertheless, AHP-based suitability remains sensitive

to the chosen factors and weights, and an area-level suitability class does not confirm the condition, ownership, capacity, or availability of a specific building. Candidate centers in OverFlow should therefore be verified as facilities and treated as configurable resources rather than inferred solely from a suitability surface.

Yin et al. proposed a system-level method for citywide coastal-flood evacuation that gives specific attention to elderly residents with special needs [16]. Their comparison of Shanghai and New York City combined exposed population, evacuation zones, shelter locations and capacities, and origin-to-destination transfers. Shanghai exhibited a shelter-capacity shortfall of approximately 230,000 spaces for the modeled conditions, while the potential demand among vulnerable elderly residents was approximately 520,000. The authors found that risk-informed shelter deployment could meet this demand and improve transfer efficiency by a factor of three.

The study shows that aggregate evacuation planning can obscure the needs of residents who require additional time, assistance, or specialized transport. It also demonstrates that shelter sufficiency depends on spatial distribution and transfer feasibility rather than citywide capacity alone. OverFlow can reflect this principle by keeping affected-population groups, center capacities, and resource requirements visible in each scenario. However, the project cannot claim equivalent improvements without local demographic, mobility, capacity, and transport data, and its initial allocation method must state how vulnerable groups are represented.

Janatabadi and Ermagun combined coastal flood risk and emergency-shelter access in a national assessment of communities across 27 coastal states in the United States [17]. Their Flood-Access Integrated Risk approach used census block groups, road-network travel, shelter locations, flood-risk indicators, and bivariate spatial clustering. Of roughly 80,000

square miles classified as high flood risk, 98% had low shelter access, and 17.8 million people lived in the resulting priority areas. Racial and ethnic minorities, older adults, people with disabilities, and households with mobility constraints were disproportionately represented.

This evidence broadens shelter evaluation from capacity and distance to accessibility and equity. A route may be computationally feasible while remaining impractical for residents without private vehicles or with limited mobility. For OverFlow, the study supports reporting uncovered demand and access differences instead of presenting a single route set as equally usable by all residents. Its national indicators and 30-minute driving threshold are context-specific, so a Philippine deployment would require locally appropriate population categories, transport assumptions, and accessibility limits.

Chen et al. addressed shelter capacity through an urban-scale evacuation simulation and spatial optimization model [18]. They constructed an origin-destination cost matrix from population-demand points, available shelters, and the road network, then assigned evacuees while enforcing shelter capacities. Remaining evacuees who could not be accommodated were redistributed to nearby plots where new fixed or temporary shelter space could be developed. In their application, 680 of 2,334 demand points could not be completely evacuated through the existing arrangement, and only 218 of 888 fixed shelters remained below capacity. The optimization consequently recommended 487 additional fixed shelters and 360 temporary shelters across the analyzed districts.

Although the model is designed for general urban emergency-shelter planning rather than a specific flood event, it provides direct evidence that aggregate citywide capacity can conceal local service gaps. It also offers a useful method for recording unassigned demand instead of forcing every population group into an already full center. OverFlow

can adopt this transparency by reporting unmet demand, remaining center capacity, and alternative destinations for each scenario. The scale of Chen et al.'s recommendations is specific to their case, and their model does not establish the safety of routes under changing flood conditions, so capacity allocation in OverFlow must remain coupled with flood-aware network analysis.

2.4 Evacuation Simulation and Related Systems

Aketa et al. developed a GIS-based evacuation action analysis system that links flood analysis with network analysis for multiple evacuees and shelters [19]. The system simulates evacuation movement, considers walking speed, and searches routes across a road network to support route and shelter planning. Although published well outside the recent review window, it is retained as an early related system that demonstrates the feasibility of coupling flood conditions with network-based evacuation behavior in one geospatial environment.

Its age and brief reported evaluation limit its value as evidence for current web architecture or modern local-government deployment. More significantly, the system emphasizes route and shelter analysis but does not establish the broader scenario controls, resource-allocation outputs, or comparative planning interface proposed for OverFlow. Its relevance is therefore conceptual: it shows that flood and evacuation analyses should interact, while the present study must establish a current implementation and evaluate whether planners can use its combined outputs effectively.

Mizumura et al. tested a routing strategy called Evacuating Inundated Areas via the Shortest Possible Route, which first minimizes movement within the inundated area before directing an evacuee toward a shelter [20]. The authors linked time-series depths from a Rainfall–Runoff–Inundation model to an Artisoc multi-agent pedestrian simulation and

used A* search on an OpenStreetMap-derived road network in Kuki City, Japan. Across the tested rainfall and departure conditions, the method reduced distance traveled on flooded roads and increased travel on safer road segments compared with ordinary shortest-path routing.

The study demonstrates that minimizing total distance and minimizing exposure to floodwater are different routing objectives. The safer route could also increase total travel distance or time for some evacuees, so the method was not uniformly preferable under every condition. This trade-off is directly relevant to OverFlow: route comparisons should disclose distance, estimated time, and flood exposure rather than label the shortest path as inherently safest. Because Mizumura et al. used hydraulic depth estimates and a calibrated simulation environment, OverFlow must not imply the same safety precision from terrain-based accumulation alone.

The United States Army Corps of Engineers' LifeSim 2.0 is a more comprehensive related system for simulating population redistribution, warnings, evacuation traffic, flood exposure, and potential life loss [21]. Its agent-based model represents interactions among evacuees, roads, vehicles, and an advancing hazard; it can reroute groups when roads flood and uses repeated simulations to represent uncertainty. LifeSim demonstrates the analytical value of treating evacuation as a dynamic process and of comparing alternative warning, road, and hazard scenarios rather than relying on a single deterministic result.

LifeSim also clarifies the intended boundary of OverFlow. Reproducing an agent-based traffic and life-loss model would require behavioral, vehicle, warning, hazard-intensity, and validation data beyond the confirmed scope of the project. OverFlow can apply the more limited but relevant principle of configurable scenario comparison without presenting its terrain-based accumulation output as a calibrated hydraulic forecast or its planning estimates

as life-loss predictions. LifeSim therefore serves both as a design reference and as evidence for clearly documenting model assumptions and uncertainty.

González-Villa et al. presented an evacuation management system that combines a real-time GIS decision-support interface, damaged-area information, shelter and assembly-point routing, and stochastic pedestrian and vehicle models [22]. In a case based on the 2019 Gran Canaria wildfire, the system handled 13 urban areas and four simultaneous evacuations and satisfied the three operational requirements defined for the demonstration. Processing demand was influenced mainly by the number of evacuees and their interactions on roads rather than by the number of routes alone.

Although the case concerns wildfire rather than flooding, the system is relevant as an architectural comparison for coordinating scenario data, network analysis, simulation, and map-based decisions in one application. Its demonstration also indicates that an interactive tool must balance analytical detail with response time as the affected population grows. The evaluation was functional and simulation-based, and observed evacuation times were unavailable, so it does not establish real-world outcome accuracy. For OverFlow, it supports measuring usability and processing performance separately from the validity of flood and evacuation estimates.

Du et al. developed an integrated socio-hydrological framework that couples the MIKE 21 hydrodynamic model, an agent-based model, and the MATSim large-scale traffic simulator [23]. Applied to a simulated 100-year flood in China's Xiong'an New Area, the framework tested all 31 combinations of five possible shelters, different levels of variation in household preparation time, and both individual shortest-route and system-level optimized routing. The flood simulation inundated 66.5% of the study area and affected an estimated 508,986 residents. The experiments showed that shelter location and distribution

mattered more than simply increasing the number of shelters; after three suitably distributed shelters, additional sites produced only marginal reductions in average evacuation time.

The route experiment is especially relevant to the interpretation of OverFlow's outputs. System-level routing reduced modeled clearance time from 27.5 to 15.5 hours and reduced the share of agents on the road at the traffic peak from 60.4% to 43.1%, showing that individually shortest routes may collectively create congestion. The integrated framework is substantially more data- and computation-intensive than OverFlow's confirmed scope, and it does not enforce shelter capacity in the reported shelter experiment. It nevertheless supports comparing complete scenarios at the system level and cautions against labeling independent shortest paths as the best evacuation plan for the population as a whole.

Butters and Dawson adapted an agent-based flood-evacuation model for Kibera, Kenya, using open building, population, road, footpath, flood, and shelter data [24]. They compared shelter locations, river crossings, response conditions, and population samples through repeated simulations. Shelters in the northwestern, northern, and northeastern parts of the study area performed best, while a southern exit, bridge, and riverside path repeatedly developed congestion. The authors also found that a 1% population sample produced unstable estimates of agents at risk, whereas a 10% sample was more informative for the tested model.

This application shows that useful evacuation experiments can be assembled in a data-constrained settlement with open sources, which is relevant to the practical setting of OverFlow. At the same time, its sampling result and acknowledged data limitations show that accessible data do not automatically produce reliable estimates. OverFlow should therefore document the completeness and resolution of its population, road, terrain, and center inputs, repeat scenarios where stochastic variation is present, and avoid treating

uncalibrated model outputs as precise predictions of community behavior.

Yang et al. developed an agent-based Dynamic Coastal Flood Evacuation model that couples high-resolution inundation, human response, GIS transportation networks, infrastructure failure, and shelter systems [25]. In a Shanghai scenario based on a 1,000-year storm event, only 24.2% of the affected population evacuated within 24 hours without early warning. A 24-hour warning allowed the same number of people to evacuate within nine hours and kept approximately 28% from being overtaken by inundation, yet the overall success rate increased by only 11.2% because shelter capacity remained restrictive. Optimized decisions approximately doubled modeled evacuation efficiency.

These results show that advance warning alone cannot overcome inadequate destination capacity or unequal shelter distribution. They also show the analytical value of allowing hazard progression, road availability, behavior, and facility limits to interact during a scenario. OverFlow can use warning time, center status, road closures, and affected population as comparative planning inputs, but its planned D8-based terrain analysis is less detailed than the two-dimensional inundation model used by Yang et al. Consequently, OverFlow's results should be presented as scenario comparisons rather than forecasts of actual evacuation success.

Bustami et al. used AnyLogic to construct a micro-scale agent-based evacuation model for compound wind and storm-surge hazards in Westbrook, Connecticut [26]. Household and vehicle characteristics, survey-based evacuation behavior, shelter capacities, a GIS road network, surge-related road closures, and stochastic wind disruptions were combined in hurricane scenarios. Under the most demanding compound conditions, townwide clearance increased from approximately 6.3 hours in the baseline Category 3–4 scenario to nearly 19 hours, while failure exceeded 90% in several population groups. Adding one shelter in a

highly constrained block group reduced its modeled failure rate from 90.7% to 63.3% and its clearance time from 19.6 to 3.26 hours.

The results illustrate how interacting hazards and uneven local access can produce severe neighborhood differences that citywide averages conceal. They also show how simulation can test a specific intervention, such as adding a shelter, before resources are committed. This model uses detailed behavioral, demographic, hydraulic, infrastructure, and stochastic inputs beyond OverFlow's present scope, and its percentages are scenario outputs rather than transferable forecasts. For OverFlow, the methodological lesson is to compare interventions at both overall and local levels and to state which hazards and behaviors the model does and does not represent.

Santana-Robles et al. linked evacuation planning with post-disaster aid distribution in their study of the 2021 Tula River flood in Hidalgo, Mexico [27]. An integer-linear-programming model selected temporary shelters and evacuation assignments, while a vehicle-routing model evaluated the delivery of humanitarian supplies. Using road distances obtained from Google Maps, the researchers generated evacuation routes from 13 affected areas to 34 possible shelters and compared the original demand with scenarios in which demand increased or decreased by 10%. They also calculated aid-distribution costs under alternative collection strategies involving state and municipal authorities.

The study expands the meaning of evacuation planning beyond moving people to a destination: the selected centers must also be supportable after evacuees arrive. This is relevant to OverFlow's proposed resource-distribution output and its controls for changing population and resource availability. However, the model minimizes defined logistics costs and uses a case-specific transport structure; it does not simulate flood propagation or determine whether each road remains safe during travel. OverFlow should therefore present

resource allocation as a scenario-dependent planning estimate and explicitly document the objective, available units, center demand, and road constraints used to generate it.

Woo and Kang integrated shelter selection, private-car and bus evacuation, congestion-aware destination choice, and relief-supply routing in a bi-level optimization model [28]. Their Ulsan flood scenario used a dynamic virtual-link method for user-equilibrium traffic assignment and a metaheuristic evolutionary algorithm refined through shelter-addition, removal, and swap operations. Relative to manually selecting the nearest shelters, the optimized plan reduced total modeled cost by 9.4%. Ignoring congestion underestimated evacuation time by as much as 41%, while the local-search procedure improved cost efficiency by a further 7%.

This work shows that shelter placement, vehicle availability, traffic, and post-arrival supply cannot always be optimized independently. It is particularly relevant to residents who depend on public or designated transport, since the sensitivity analysis found that smaller bus fleets increased clearance time, workload, and required shelter capacity. The model nevertheless depends on detailed demand, traffic-capacity, fleet, cost, and logistics data that are not yet confirmed for OverFlow. Its appropriate contribution is to guide the project's scenario variables and evaluation measures rather than to justify implementing the complete bi-level model.

Insani et al. formulated an integrated vehicle-routing model for evacuating vulnerable residents and delivering relief supplies before floodwater reaches the affected area [29]. The model permits vehicle reuse, multiple trips, split deliveries, uncertain evacuation demand, and closing time windows. An exact solver and a modified genetic algorithm were evaluated through synthetic instances and a case study in Bontoala, Indonesia. The integrated approach required fewer vehicles than comparable separated operations, while

the genetic algorithm produced optimal or near-optimal solutions for smaller instances approximately 92.5% faster than the exact method and remained usable when the exact approach became impractical for larger cases.

The study supports considering evacuation and relief distribution as linked uses of limited vehicles and time. It also demonstrates why a practical decision-support system may need a well-evaluated heuristic when exact optimization cannot respond quickly enough. However, the model addresses the early stage before inundation reaches demand points and relies on assumptions about demand, time windows, and vehicle operation. OverFlow should adopt these concepts only if the required local data and objectives are confirmed; otherwise, it should report resource allocation as a bounded scenario estimate and retain unmet needs explicitly.

2.5 Terrain Processing and Evacuation Behavior

O'Callaghan and Mark introduced the method now commonly called D8, in which each cell in a digital elevation model directs flow toward its steepest downslope neighbor, allowing drainage direction and accumulated contributing area to be derived from gridded terrain [30]. This is a classic methodological source rather than a recent evacuation study, and it provides the foundation for OverFlow's confirmed terrain-based accumulation approach. D8 offers a computationally straightforward way to identify likely flow paths and areas of accumulation from elevation data.

The method does not by itself calculate flood depth, velocity, drainage-system behavior, or the timing of inundation. Consequently, terrain-derived accumulation should be used in OverFlow to classify scenario-specific flood-affected areas according to an explicitly documented threshold and should not be described as a calibrated hydraulic model. This

distinction affects the methodology: the study must validate the digital elevation model, document preprocessing and threshold choices, and test how the resulting flood classification changes the set of usable road links.

Lim et al. examined the route choices of households affected by flooding in Bagong Silangan, Quezon City, using three binary logit models based on hazard, socioeconomic, and evacuation-related variables [31]. Their Philippine case shows that evacuees do not necessarily select routes solely by minimum distance; departure timing, familiarity, perceived conditions, and household characteristics can influence movement. The study provides local evidence that a mathematically shortest route and an actual human route are not always equivalent.

This finding is critical for interpreting OverFlow's outputs. A feasible or recommended path should be presented as a planning option derived from the available network and scenario constraints, not as a prediction of the route every resident will follow. The study also identifies a methodological opportunity for evaluation: proposed routes can be reviewed by local disaster personnel and, where data are available, compared with known routes, accessibility conditions, and community practices. Behavioral modeling may remain outside the initial implementation, but its absence should be stated as a limitation when evacuation-time estimates are interpreted.

Borowska-Stefańska et al. examined how self-evacuation from flood-hazard areas changes regional traffic conditions in Greater Poland [32]. They surveyed 500 residents about evacuation knowledge, transport choice, ability to self-evacuate, and mobility limitations, then incorporated the responses into six scenarios using a calibrated multimodal regional traffic model and the RoadLoad application. The scenarios represented different stages of planned and emergency self-evacuation, including road-management measures and

road sections made unavailable by flooding. This combination of household evidence and network simulation allowed the authors to compare flood evacuation traffic with ordinary regional travel.

The simulations found that self-evacuation substantially redistributed traffic across the road network, including areas beyond the immediate flood zone, while its effect on average speed was smaller because the modeled network retained spare capacity. Delayed and spontaneous evacuation produced the most difficult conditions, and the authors emphasized staged evacuation and current route information as planning measures. These findings reinforce Lim et al.'s evidence that evacuation movement depends on behavior and context rather than geometric distance alone. For OverFlow, manual closures and demand changes can support useful pre-event comparisons, but reliable congestion or clearance-time claims would require locally calibrated traffic demand, departure behavior, vehicle availability, and road-capacity data.

2.6 Synthesis and Research Gap

The reviewed literature shows that effective flood evacuation planning requires the integration of hazard information, road accessibility, evacuation-center suitability, population demand, and available resources. Flooding can disconnect roads, increase travel time, change the nearest reachable center, and make normally suitable routes unsafe [2, 3, 4, 6]. Recent routing studies further demonstrate that minimum distance alone is an insufficient measure of route quality. Routes may need to minimize flood exposure, avoid areas where inundation is expanding, or be recalculated when roads and destinations become unavailable [7, 8, 20]. These findings support OverFlow's objective of applying the simulated flood condition to the road network before generating evacuation routes.

The literature also establishes that evacuation-center planning cannot be reduced to selecting the nearest facility. Center safety, capacity, accessibility, population coverage, travel time, road risk, and the needs of vulnerable groups must be considered together [10, 13, 14, 18, 16, 17]. Studies that combine shelter selection with evacuation simulation found that better facility placement and assignment can improve modeled population coverage, reduce exposure, and decrease evacuation time [9, 11]. The Philippine location-routing study by de Veluz et al. further demonstrates that facility cost, travel time, vehicle availability, and center capacity produce competing planning objectives [12]. Therefore, OverFlow should present center assignments as constrained planning alternatives and explicitly report remaining capacity, unmet demand, and the assumptions used to generate each allocation.

Simulation-based studies provide evidence that evacuation plans can be examined before an actual disaster occurs. They allow researchers and planners to compare warning times, shelter arrangements, route strategies, road closures, population behavior, and resource interventions under controlled conditions [21, 23, 25, 24, 26]. Some simulations reported fewer modeled casualties, shorter clearance times, or higher evacuation success after changes to shelters or routing strategies [9, 23, 26]. However, these results are conditional on each model's hazard representation, behavioral assumptions, population data, and study area. The literature therefore supports simulation as a planning and comparison tool, but it does not establish that simulation alone will reduce casualties in an actual event. Real-world benefits depend on the accuracy of the inputs, the implementation of the resulting plan, community response, and validation with local disaster-management personnel.

Existing related systems also demonstrate the practicality of combining GIS, network analysis, and evacuation information in an interactive application [19, 2, 5, 8, 22]. These systems can display hazards, identify affected infrastructure, calculate or recalculate routes,

and support decisions without requiring users to operate specialist GIS software. Nevertheless, many systems concentrate on one part of the problem, such as road accessibility, shelter suitability, route guidance, or detailed agent-based evacuation. The more comprehensive systems require hydraulic models, behavioral surveys, traffic calibration, and extensive computing resources that may be difficult to obtain or maintain in a local-government setting.

The Philippine studies provide relevant evidence but remain divided across separate planning concerns. Lim et al. examined the behavioral factors affecting route choice in Quezon City [31]; Bolanio et al. assessed evacuation-center suitability and developed a web map for Las Nieves [5]; Abuzo et al. compared road conditions under normal and flooded scenarios in Cagayan de Oro [6]; and de Veluz et al. optimized facilities, routes, and vehicles at the provincial scale [12]. Together, these studies establish the local relevance of flood-aware routing, center assessment, scenario comparison, and resource planning. However, none of the reviewed Philippine studies provides a single accessible system that combines terrain-derived flood scenarios, affected-road identification, capacity-constrained center assignment, route comparison, and resource estimates within one localized planning workflow.

This separation defines the research gap addressed by OverFlow. There remains a need for a localized, web-based decision-support system that allows disaster planners to configure flood and operational conditions, observe how those conditions affect the road network, compare feasible evacuation routes, assign affected populations according to center accessibility and capacity, and examine resource requirements across alternative scenarios. OverFlow addresses this gap through an integrated and comparatively lightweight planning environment based on locally obtainable terrain, road, population, center, and resource data.

The study does not attempt to reproduce a calibrated hydraulic, traffic, behavioral, or life-loss model. Its terrain-based D8 process identifies drainage direction and relative accumulation rather than verified flood depth or timing [30]. Consequently, the methodology must separately evaluate the flood classification, routing and allocation feasibility, scenario sensitivity, processing performance, and usefulness of the interface to intended planners. This evaluation will determine whether OverFlow provides a credible planning aid while keeping its outputs within the limits of the available data and computational approach.

3 Methodology

3.1 Research Design

3.2 Data Gathering Tools

3.3 Research Instruments

3.4 Data Gathering Procedures

3.5 Statistical Treatment of Data

4 Results and Discussion

4.1 System Development and Scenario-Analysis Results

4.2 System Evaluation Results

5 Conclusions and Recommendations

5.1 Conclusion

5.2 Recommendation

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APPENDICES

A Detailed Tables

B Questionnaires